

Robot Dog Mechanical Design

Initial Mechanical Design Report

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Smart Methods Summer Training 2026

1. Introduction

This project presents the initial mechanical design of a quadruped robot dog. The design focuses on developing a lightweight, stable, and simple mechanical structure. It includes the body frame, leg design, joints, motor selection, torque estimation, stability analysis, and walking method.

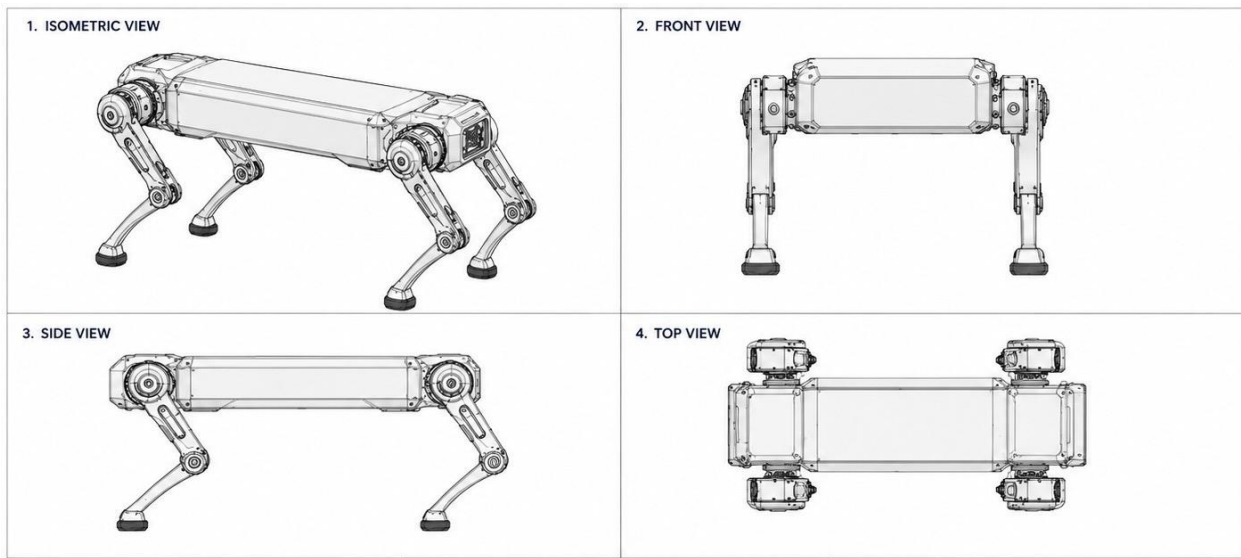


Figure 1. Robot Body Views (Isometric, Front, Side, and Top)

2. Overall Body and Frame Design

The robot uses a lightweight aluminum alloy frame with a symmetrical body layout. The internal space is designed to accommodate the battery, controller, and wiring while maintaining good weight distribution.

Parameter	Value
Robot Type	Quadruped Robot
Body Material	Aluminum Alloy
Body Length	35 cm
Body Width	18 cm
Body Height	12 cm

Design Features

- Lightweight frame
- High structural strength
- Symmetrical body
- Space for electronics
- Easy future upgrades

3. Leg Design

Each leg consists of:

- Hip Joint
- Upper Leg
- Knee Joint
- Lower Leg
- Foot

The two-joint design provides enough flexibility while keeping the mechanical structure simple and lightweight.

4. Joints and Degrees of Freedom (DOF)

Each leg has:

- 1 Hip Joint
- 1 Knee Joint

Total DOF per leg = 2

Parameter	Value
Estimated Weight	6 kg
Degrees of Freedom	8 DOF

Total DOF = 8

5. Motor Selection

The robot uses Servo Motors because they provide:

- High positioning accuracy
- High torque
- Compact size
- Easy control
- Reliable performance

6. Preliminary Torque Calculation

Assumptions

- Leg Mass = 0.4 kg
- Leg Length = 0.15 m
- Gravity = 9.81 m/s^2

Force:

$$F = m \times g$$

$$F = 0.4 \times 9.81 = 3.92 \text{ N}$$

Torque:

$$T = F \times d$$

$$T = 3.92 \times 0.15 = 0.59 \text{ N}\cdot\text{m}$$

Recommended Servo Torque $\geq 0.6 \text{ N}\cdot\text{m}$

7. Stability and Center of Gravity

The battery and controller are placed near the center of the robot body to maintain a low center of gravity.

- Symmetrical body
- Central battery location
- Wide support base
- Low center of gravity

8. Walking Method

The selected gait is Trot Gait.

1. Front Left + Rear Right
2. Front Right + Rear Left

This gait provides good balance, smooth movement, and efficient energy consumption.

9. Expected Mechanical Challenges

- Limited motor torque
- Joint friction
- Foot slippage
- Structural vibration
- Uneven weight distribution
- Mechanical wear

10. Conclusion

The proposed robot dog is a lightweight quadruped prototype designed with simplicity, stability, and future development in mind. The design provides a strong foundation for future improvements, including sensor integration and autonomous navigation.